

# RA Concept Questions Review

## 中文讲解与英文原题整理

Kinematics, Dynamics and Control of Robots

### 使用说明

这份资料把 RA quiz model answers 中和概念、定义、公式适用条件有关的题目整理在一起。每题保留英文题干或英文选项，并用中文解释为什么对、为什么错。错误命题已经改成正确说法。纯粹的 “What was most difficult” 自由回答题没有收录；部分数值题只保留其核心公式和正确数值，方便复习时检查方法。

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# 1 March 16 Lecture: Hat, Vee and Twist Basics

## 1.1 Hat Map and Cross Product Identity

**English statement:** Consider two column vectors  $a \in \mathbb{R}^3$  and  $b \in \mathbb{R}^3$ . Indicate whether the following is true or false:

$$\hat{a}b = \hat{b}^T a.$$

**Answer:** True.

**中文讲解:** 三维向量  $a$  的 hat map 是把向量变成一个 skew-symmetric matrix, 使得矩阵乘法等价于叉乘:

$$\hat{a}b = a \times b.$$

因为叉乘反交换:

$$a \times b = -b \times a.$$

而  $\hat{b}^T = -\hat{b}$ , 所以

$$\hat{b}^T a = -\hat{b}a = -(b \times a) = a \times b = \hat{a}b.$$

因此这个等式是对的。这里最容易忘的是:  $\hat{b}^T$  不是另一个新的运算, 它只是  $\hat{b}$  的转置, 而 skew-symmetric 的转置等于负号。

**关键公式:**

$$\hat{a} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}, \quad \hat{a}b = a \times b, \quad \hat{a}^T = -\hat{a}.$$

## 1.2 Vee Map of a Twist Matrix

**English statement:** Consider the scalar numbers  $a, b, c, d, e, f$  and  $g$ . Indicate whether the following is true or false:

$$\begin{bmatrix} 0 & a & b & d \\ -a & 0 & c & e \\ -b & -c & 0 & f \\ 0 & 0 & 0 & 0 \end{bmatrix}^\vee = \begin{bmatrix} d \\ e \\ f \\ -c \\ b \\ -a \end{bmatrix}.$$

**Answer:** True.

**中文讲解:** MLS 里常用的 twist column vector 顺序是

$$\xi = \begin{bmatrix} v \\ \omega \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}.$$

它的  $4 \times 4$  matrix form 是

$$\hat{\xi} = \begin{bmatrix} \hat{\omega} & v \\ 0 & 0 \end{bmatrix}.$$

所以反 hat，也就是 vee map，要先从第 4 列取出  $v = [d, e, f]^T$ ，再从左上角  $3 \times 3$  skew block 读出  $\omega$ 。题目左上角矩阵满足

$$\hat{\omega} = \begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}.$$

逐项比较得到

$$\omega_1 = -c, \quad \omega_2 = b, \quad \omega_3 = -a.$$

因此

$$\hat{\xi}^v = \begin{bmatrix} v \\ \omega \end{bmatrix} = \begin{bmatrix} d \\ e \\ f \\ -c \\ b \\ -a \end{bmatrix}.$$

### 1.3 Is the $4 \times 4$ Twist Matrix Skew-Symmetric?

**English statement:** The  $4 \times 4$  twist matrix  $\hat{\xi}$  is a skew-symmetric matrix.

**Answer:** False.

**Corrected statement:** The upper-left  $3 \times 3$  block  $\hat{\omega}$  of  $\hat{\xi}$  is skew-symmetric, but the full  $4 \times 4$  twist matrix is generally not skew-symmetric.

**中文讲解:** 对一个 revolute 或 general twist,

$$\hat{\xi} = \begin{bmatrix} \hat{\omega} & v \\ 0 & 0 \end{bmatrix}.$$

虽然  $\hat{\omega}^T = -\hat{\omega}$ ，但是整个  $4 \times 4$  矩阵的第 4 列一般是  $[v^T, 0]^T$ ，第 4 行却是  $[0, 0, 0, 0]$ 。如果整个矩阵要 skew-symmetric，就必须满足第 4 行等于第 4 列转置取负，这通常不成立，除非  $v = 0$ 。所以题目说整个 twist matrix 都是 skew-symmetric 是错的。

### 1.4 Fixed-Axis Rotation Velocity

**English statement:** MLS equation (2.25)

$$\dot{p}(t) = \hat{\omega}(p(t) - q)$$

is based on the fact that the velocity vector of a point on a rigid body undergoing fixed-axis rotation is the cross product of the angular velocity vector of the body with the position vector of the point in question, relative to a point on the axis of rotation.

**Answer:** True.

**中文讲解:** 如果刚体绕一条固定轴转动，轴的方向是单位向量  $\omega$ ，轴上一点是  $q$ 。刚体上一点  $p(t)$  的速度由角速度和相对位置叉乘给出：

$$\dot{p}(t) = \omega \times (p(t) - q).$$

用 hat map 把叉乘写成矩阵乘法，就得到

$$\dot{p}(t) = \hat{\omega}(p(t) - q).$$

这里一定要注意相对位置是  $p(t) - q$ ，不是直接用  $p(t)$ 。因为绕轴转动的速度取决于点到旋转轴的相对位置。

## 2 March 23 Lecture: Body Velocity, Spatial Velocity and Adjoint

### 2.1 Rotation Matrix Derivative

**English statement:** Consider the rotation matrix  $R$  that describes the rotation of a coordinate frame  $B$  relative to an inertial frame  $A$ . Indicate which option is the best partial description of the result when the first time derivative of  $R$  is pre-multiplied by  $R^T$ .

**Answer:** Correct option: a  $3 \times 3$  skew symmetric matrix with the components in frame  $B$  of the angular velocity vector of frame  $B$ .

**中文讲解:** 题目问的是

$$R^T \dot{R}.$$

因为  $R \in SO(3)$ ，所以

$$R^T R = I.$$

两边对时间求导：

$$\dot{R}^T R + R^T \dot{R} = 0.$$

这说明  $R^T \dot{R}$  的转置等于它的负号，所以它是 skew-symmetric matrix。它对应的是 angular velocity expressed in body frame，也就是

$$\hat{\omega}^b = R^T \dot{R}.$$

对比地， $\dot{R} R^T$  才是 spatial frame 里的 angular velocity：

$$\hat{\omega}^s = \dot{R} R^T.$$

**关键公式：**

$$\hat{\omega}^b = R^T \dot{R}, \quad \hat{\omega}^s = \dot{R} R^T, \quad \omega^s = R \omega^b.$$

### 2.2 Rigid Body Motion Derivative

**English statement:** Consider a rigid body motion  $g_{ab}(t) \in SE(3)$ . Indicate which option is the best partial description of the result when the first time derivative of  $g_{ab}(t)$  is pre-multiplied by the inverse of  $g_{ab}(t)$ .

**Answer:** Correct option: A twist in  $4 \times 4$  matrix form, of which the first three rows of the 4th column are the components of the velocity, in the body fixed frame, of the origin of this frame, relative to the spatial frame.

**中文讲解:** 题目问的是

$$g_{ab}^{-1} \dot{g}_{ab}.$$

这就是 body velocity 的 matrix form:

$$\hat{V}^b = g^{-1}\dot{g}.$$

它是一个  $4 \times 4$  twist matrix:

$$\hat{V}^b = \begin{bmatrix} \hat{\omega}^b & v^b \\ 0 & 0 \end{bmatrix}.$$

第 4 列前三个元素是  $v^b$ , 也就是 velocity part expressed in body fixed frame。注意它不是 spatial frame 的速度分量; spatial velocity 的 matrix form 是

$$\hat{V}^s = \dot{g}g^{-1}.$$

## 2.3 Adjoint Transformation Size

**English statement:** The adjoint transformation associated with the rigid motion  $g_{ab}(t)$  is a  $6 \times 6$  matrix.

**Answer:** True.

**中文讲解:** twist 是 6 维向量:

$$V = \begin{bmatrix} v \\ \omega \end{bmatrix} \in \mathbb{R}^6.$$

所以把一个 twist 从 body frame 变到 spatial frame 的 adjoint transformation 必须是  $6 \times 6$  矩阵。若

$$g = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix},$$

在 MLS 的  $[v; \omega]$  顺序下,

$$\text{Ad}_g = \begin{bmatrix} R & \hat{p}R \\ 0 & R \end{bmatrix}.$$

## 2.4 Spatial Twist from Body Twist

**English statement:** The velocity of a body expressed as a twist in the spatial frame can be calculated by post-multiplying the adjoint transformation associated with the rigid body motion by the velocity of the body expressed as a twist in the body fixed frame.

**Answer:** True.

**中文讲解:** 这里的 “post-multiplying” 指的是矩阵在左边、向量在右边:

$$V^s = \text{Ad}_g V^b.$$

这不是说两种速度是不同物理运动; 它们描述的是同一个 rigid body velocity, 只是分量表达在不同坐标系中。spatial twist 是 expressed in spatial frame, body twist 是 expressed in body fixed frame。

**关键公式:**

$$\hat{V}^s = \dot{g}g^{-1}, \quad \hat{V}^b = g^{-1}\dot{g}, \quad V^s = \text{Ad}_g V^b.$$

### 3 April 13 Lectures: Wrench as a Screw

#### 3.1 Pitch of a Wrench

**English statement:** The net wrench applied to an object is

$$F = \begin{bmatrix} 1.2 & 0.4 & 0.8 & 0.15 & 0.21 & 0.08 \end{bmatrix}^T.$$

Calculate the pitch of this wrench as a screw.

**Answer:** Correct value: 0.14643.

**中文讲解:** 把 wrench 写成 force part 和 moment/torque part:

$$F = \begin{bmatrix} f \\ \tau \end{bmatrix}, \quad f = \begin{bmatrix} 1.2 \\ 0.4 \\ 0.8 \end{bmatrix}, \quad \tau = \begin{bmatrix} 0.15 \\ 0.21 \\ 0.08 \end{bmatrix}.$$

当 force part 非零时, wrench 作为 screw 的 pitch 是 moment 在 force 方向上的投影除以 force magnitude squared:

$$h = \frac{f^T \tau}{\|f\|^2}.$$

代入:

$$f^T \tau = 1.2(0.15) + 0.4(0.21) + 0.8(0.08) = 0.328,$$

$$\|f\|^2 = 1.2^2 + 0.4^2 + 0.8^2 = 2.24.$$

所以

$$h = \frac{0.328}{2.24} = 0.14642857 \approx 0.14643.$$

#### 3.2 Magnitude of a Wrench

**English statement:** The net wrench applied to an object is

$$F = \begin{bmatrix} 1.2 & 0.4 & 0.8 & 0.15 & 0.21 & 0.08 \end{bmatrix}^T.$$

Calculate the magnitude of this wrench as a screw.

**Answer:** Correct value: 1.4967.

**中文讲解:** 对于 force part 非零的 wrench screw, magnitude 通常取 force part 的长度:

$$\|f\| = \sqrt{1.2^2 + 0.4^2 + 0.8^2} = \sqrt{2.24} = 1.49666 \dots$$

所以正确答案是

$$1.4967.$$

pitch 描述“沿 screw axis 每单位转动对应的平移比例”或 wrench 中 torque 沿 force 方向的比例; magnitude 描述 screw 的主强度, 这里就是 force magnitude.

## 4 April 20 Lectures: Product of Exponentials Forward Kinematics

### 4.1 Tool Origin Coordinates from Forward Kinematics

**English statement:** Consider a robotic manipulator in a new configuration. You need to report the coordinates, in mm, of the origin of the tool frame in this new configuration, expressed in the spatial frame. Correct answers:  $x = -1202.64$ ,  $y = 1238.97$ ,  $z = 770$ .

**Answer:** This is a forward kinematics question.

**中文讲解:** 这类题不是问 inverse kinematics, 而是: 已知每个 joint 的位移或角度, 求 tool frame 在 spatial frame 中的位置。核心是 product of exponentials formula:

$$g_{st}(\theta) = e^{\hat{\xi}_1 \theta_1} e^{\hat{\xi}_2 \theta_2} \dots e^{\hat{\xi}_n \theta_n} g_{st}(0).$$

如果最后得到

$$g_{st}(\theta) = \begin{bmatrix} R(\theta) & p(\theta) \\ 0 & 1 \end{bmatrix},$$

那么 tool frame origin expressed in spatial frame 就是第 4 列前三个元素:

$$p(\theta) = g_{st}(1 : 3, 4).$$

题目给出的三个正确数值只是  $p(\theta)$  的三个坐标:

$$p(\theta) = \begin{bmatrix} -1202.64 \\ 1238.97 \\ 770 \end{bmatrix} \text{ mm.}$$

**关键公式:** Revolute joint 的 exponential 是

$$e^{\hat{\xi}\theta} = \begin{bmatrix} e^{\hat{\omega}\theta} & (I - e^{\hat{\omega}\theta})(\omega \times v) + \omega \omega^T v \theta \\ 0 & 1 \end{bmatrix},$$

其中  $\xi = [v; \omega]$ 。Prismatic joint 的 exponential 是

$$e^{\hat{\xi}\theta} = \begin{bmatrix} I & v\theta \\ 0 & 1 \end{bmatrix}, \quad \omega = 0.$$

## 5 April 27 Lectures: Zero-Pitch Twist and Subproblem 1

### 5.1 Rotation Angle from a Zero-Pitch Unit Twist

**English statement:** A zero pitch unit twist is defined by

$$\xi = \begin{bmatrix} 3.2733 \\ 2.4004 \\ -3.0551 \\ 0.4364 \\ -0.8729 \\ -0.2182 \end{bmatrix}.$$



The point with coordinates  $(3, 1, 4)$  lies on the twist axis. The point with coordinates  $(4.7808, 1.9852, -2.2792)$  is generated with a rigid body rotation defined by this twist of the point with coordinates  $(1.2, 0.2, -2.3)$ . Determine the angle of this rotation.

**Answer:** Correct value:  $36^\circ$ .

**中文讲解:** “zero pitch unit twist” 表示这是纯转动，不含沿轴方向的螺旋平移。twist 写成

$$\xi = \begin{bmatrix} v \\ \omega \end{bmatrix}, \quad \|\omega\| = 1, \quad v = -\omega \times q$$

其中  $q$  是旋转轴上一点。题目已经给出轴上一点

$$q = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}.$$

令原点为

$$p = \begin{bmatrix} 1.2 \\ 0.2 \\ -2.3 \end{bmatrix},$$

旋转后的目标点为

$$r = \begin{bmatrix} 4.7808 \\ 1.9852 \\ -2.2792 \end{bmatrix}.$$

把点都减去轴上一点  $q$ ，得到从旋转轴看出来的两个向量：

$$u = p - q, \quad w = r - q.$$

若没有数值误差， $u$  和  $w$  到轴的垂直距离相同。旋转角可用  $\text{atan2}$  判定符号：

$$\theta = \text{atan2}(\omega^T(u \times w), u^T w).$$

这个题算出来是

$$\theta = 36^\circ.$$

这就是 inverse kinematics 里 Subproblem 1 的基本形式：已知一条旋转轴和点  $p$ ，求转多少角能把  $p$  转到目标点。

## 6 May 11 Lectures: Spatial Jacobian and Wrench Matching

### 6.1 Benefit of Product of Exponentials

**English statement:** One of the benefits of using the product of exponentials formula to present the forward kinematics of a manipulator is that the spatial Jacobian of the manipulator can be determined almost by inspection, by filling in each column as the corresponding joint twist transformed to its current manipulator configuration.

**Answer:** True.

**中文讲解:** POE 形式的正向运动学是

$$g(\theta) = e^{\hat{\xi}_1 \theta_1} e^{\hat{\xi}_2 \theta_2} \dots e^{\hat{\xi}_n \theta_n} g(0).$$

spatial Jacobian 的第  $i$  列不是简单等于  $\xi_i$ , 而是前面所有 joint 已经运动后, 把第  $i$  个 twist 变换到当前 spatial 表达:

$$J_i^s(\theta) = \text{Ad}_{e^{\hat{\xi}_1 \theta_1} \dots e^{\hat{\xi}_{i-1} \theta_{i-1}}} \xi_i.$$

所以每一列都可以从 POE 结构 “almost by inspection” 写出来。

**关键公式:**

$$V_{st}^s = J_s(\theta) \dot{\theta}.$$

## 6.2 Tool Frame Instantaneous Velocity

**English statement:** For the manipulator in the quiz, the correct 1st element of the instantaneous velocity of the tool frame, in components in the spatial frame, is 0.401452 m/s. The correct last element is 0.45 rad/s.

**Answer:** These values come from  $V_{st}^s = J_s(\theta) \dot{\theta}$ .

**中文讲解:** 题目给 joint velocities:

$$\dot{\theta} = \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \\ \dot{\theta}_4 \end{bmatrix}.$$

只要构造出 spatial Jacobian,

$$J_s(\theta) = \begin{bmatrix} J_1^s & J_2^s & J_3^s & J_4^s \end{bmatrix},$$

tool frame 的 instantaneous spatial twist 就是

$$V_{st}^s = J_s(\theta) \dot{\theta}.$$

如果课程采用 MLS 的  $[v; \omega]$  顺序, 那么前 3 个元素是 linear velocity, 后 3 个元素是 angular velocity。quiz 中第 1 个元素是

$$0.401452 \text{ m/s},$$

最后一个 angular velocity 元素是

$$0.45 \text{ rad/s}.$$

## 6.3 Joint Torques and Wrench Coordinates

**English statement:** The joint torques vector can be calculated by postmultiplying the transpose of the spatial manipulator Jacobian with the wrench applied at the tool, expressed in the tool frame.

**Answer:** False.

**Corrected statement:** Use the spatial Jacobian with a wrench expressed in the spatial frame, or use the body Jacobian with a wrench expressed in the tool/body frame.

**中文讲解：**这个题的错误点不是  $J^T F$  这个形式，而是坐标系不匹配。虚功关系要求 twist 和 wrench 必须用同一个 frame 表达。正确写法是：

$$\tau = J_s(\theta)^T F_s,$$

其中  $F_s$  是 expressed in the spatial frame 的 wrench；或者

$$\tau = J_b(\theta)^T F_b,$$

其中  $F_b$  是 expressed in the body/tool frame 的 wrench。题目把 spatial Jacobian 和 tool-frame wrench 混在一起，所以是 False。

## 7 May 18 Lectures: Kinetic Energy and Rigid Body Inertia

### 7.1 Does Kinetic Energy Depend Only on Velocities?

**English statement:** The kinetic energy  $T$  of a mechanical system can be expressed as a summation over all the mass particles comprising the system,

$$T = \frac{1}{2} \sum_i m_i \|\dot{r}_i\|^2.$$

Statement:  $T$  is seen to be dependent on velocities only and thus it is independent of the generalized coordinates.

**Answer:** False.

**Corrected statement:** In generalized coordinates, kinetic energy generally depends on both generalized velocities and generalized coordinates.

**中文讲解：**最原始的粒子形式确实只写了每个粒子的速度：

$$T = \frac{1}{2} \sum_i m_i \|\dot{r}_i\|^2.$$

但在 analytical dynamics 里，粒子位置通常是 generalized coordinates 的函数：

$$r_i = r_i(q).$$

所以

$$\dot{r}_i = \frac{\partial r_i}{\partial q} \dot{q}.$$

代回去以后，动能变成

$$T = \frac{1}{2} \dot{q}^T M(q) \dot{q}.$$

这里  $M(q)$  会随着构型  $q$  改变，所以一般有

$$\frac{\partial T}{\partial q_i} \neq 0.$$

因此“只依赖速度、与 generalized coordinates 无关”是错误的。

## 7.2 Moment of Inertia Tensor as a Volume Integral

**English statement:** The moment of inertia tensor of a rigid body, expressed in components in the body frame with origin at the centre of mass of the body, can be calculated as the volume integral over the volume  $V$  of the body:

$$\int_V \rho(r) \hat{r}^T \hat{r} dV.$$

**Answer:** True.

**中文讲解:** 刚体的 inertia tensor 描述质量分布对转动的影响。若  $r$  是 body frame 中从质心指向体内某点的向量,  $\rho(r)$  是密度, 则

$$I = \int_V \rho(r) \hat{r}^T \hat{r} dV.$$

利用 hat map 的性质:

$$\hat{r}^T \hat{r} = \|r\|^2 I_{3 \times 3} - r r^T,$$

所以也常写成

$$I = \int_V \rho(r) (\|r\|^2 I - r r^T) dV.$$

题目说的是 body frame 且 origin 在 centre of mass, 所以这个积分表达式是正确的。

## 7.3 Rigid Body Torque Is Not Just $I\alpha$

**English statement:** The torque necessary to cause a rigid body to have an angular acceleration vector expressed as  $\alpha^b = \dot{\omega}^b$ , in components in the body frame, can be calculated, in components in the body frame, as the moment of inertia matrix calculated in components in the body frame, postmultiplied by  $\alpha^b$ .

**Answer:** False.

**Corrected statement:** In body-frame components, the rotational equation is

$$\tau^b = I\alpha^b + \dot{\omega}^b I\omega^b.$$

**中文讲解:** 如果角速度方向不变, 或者在某些非常特殊的情况下, 可能会看到类似  $\tau = I\alpha$  的简化式。但一般刚体在 body frame 中的欧拉方程是

$$\tau^b = I\dot{\omega}^b + \omega^b \times (I\omega^b).$$

用 hat map 表示叉乘:

$$\tau^b = I\alpha^b + \hat{\omega}^b I\omega^b.$$

第二项是 gyroscopic term。它来自 angular momentum 的方向变化, 不能随便丢掉。所以题目说只要  $I\alpha^b$  就能算 torque, 是错误的。

## 8 May 25 Lectures: Manipulator Inertia Matrix

### 8.1 Manipulator Kinetic Energy and Joint Angles

**English statement:** The kinetic energy of an open-chain manipulator can be expressed as half of the manipulator inertia matrix premultiplied with the transpose of the  $\dot{\theta}$  vector and postmultiplied with the  $\dot{\theta}$  vector. This means that the kinetic energy of the manipulator is independent of the joint angles  $\theta$ .

**Answer:** False.

**Corrected statement:** The kinetic energy is

$$T = \frac{1}{2} \dot{\theta}^T M(\theta) \dot{\theta},$$

and it generally depends on  $\theta$  through  $M(\theta)$ .

**中文讲解:** 前半句是对的:

$$T = \frac{1}{2} \dot{\theta}^T M(\theta) \dot{\theta}.$$

但后半句错在把  $M(\theta)$  当成常数。机械臂的每个 link 的位置和姿态都会随 joint angles 改变, Jacobian 也会随  $\theta$  改变, 因此由 link inertia 合成出来的 manipulator inertia matrix 也一般是

$$M = M(\theta).$$

所以  $T$  不只是  $\dot{\theta}$  的函数, 也通常依赖  $\theta$ 。

### 8.2 MLS Lemma 4.2: Symmetry of $M$

**English statement:** The first property listed in the statement of MLS Lemma 4.2 is used in the proof of the second.

**Answer:** True.

**中文讲解:** quiz feedback 里指出, 第二个性质证明最后一步是:

$$\text{Switching } i \text{ and } j \text{ shows that } (\dot{M} - 2C)^T = -(\dot{M} - 2C).$$

这一步成立依赖

$$M_{ij} = M_{ji},$$

也就是  $M$  是 symmetric matrix。这个 symmetry 正是 Lemma 4.2 的第一个性质的一部分。直观上,  $M(\theta)$  来自动能二次型:

$$T = \frac{1}{2} \dot{\theta}^T M(\theta) \dot{\theta},$$

而二次型中的 inertia matrix 必须是 symmetric positive definite。因此第一个性质确实被用到了第二个性质的证明中。

## 9 June 4 Lectures: Computed Torque and Workspace Control

### 9.1 Computed Torque Control

**English statement:** Indicate which option regarding the computed torque control technique is false. Correct false option: The computed torque control technique is central to workspace control.

**Answer:** False option: “The computed torque control technique is central to workspace control.”

**Corrected statement:** Computed torque control is a feedback linearization technique for robot dynamics; workspace control is about controlling the end-effector trajectory directly in task space and is not tied to one specific torque-control method.

**中文讲解:** computed torque control 的核心思想是用模型抵消非线性动力学。机械臂动力学通常写成

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + N(\theta, \dot{\theta}) = \tau.$$

computed torque control 设计 torque, 让闭环误差近似变成线性二阶系统。例如给定期望关节轨迹  $\theta_d$ , 常见形式是

$$\tau = M(\theta) \left( \ddot{\theta}_d + K_d \dot{e} + K_p e \right) + C(\theta, \dot{\theta})\dot{\theta} + N(\theta, \dot{\theta}),$$

其中

$$e = \theta_d - \theta.$$

它确实是 feedback linearization, 也可以分成 feedforward 和 feedback 部分。但 workspace control 是 task-space/end-effector control 的概念, 重点是控制  $x$ 、 $\dot{x}$  或 end-effector pose, 不等于必须使用 computed torque control。

### 9.2 Workspace Control and Inverse Kinematics

**English statement:** In applying workspace control, the end effector trajectory is controlled directly, without having to solve the inverse kinematic problem.

**Answer:** True.

**中文讲解:** workspace control 直接在 end-effector/task space 中定义误差, 例如

$$e_x = x_d - x.$$

速度层面通过 Jacobian 联系 joint velocity 和 end-effector velocity:

$$\dot{x} = J(\theta)\dot{\theta}.$$

力/力矩层面可以通过 Jacobian transpose 或 operational-space dynamics 把 task-space command 映射回 joint torques。关键思想是: 不用先完整求出一条 joint trajectory  $\theta_d(t)$  来跟踪, 也就是不需要显式解 inverse kinematics 后再做 joint-space tracking。

## 总复习索引

| 主题                       | 必须记住的公式或判断                                                                                          |
|--------------------------|-----------------------------------------------------------------------------------------------------|
| Hat map                  | $\hat{a}b = a \times b, \hat{a}^T = -\hat{a}$                                                       |
| Twist vee                | $\hat{\xi} = \begin{bmatrix} \hat{\omega} & v \\ 0 & 0 \end{bmatrix}, \hat{\xi}^\vee = [v; \omega]$ |
| Body angular velocity    | $\hat{\omega}^b = R^T \dot{R}$                                                                      |
| Spatial angular velocity | $\hat{\omega}^s = \dot{R} R^T$                                                                      |
| Body twist               | $\hat{V}^b = g^{-1} \dot{g}$                                                                        |
| Spatial twist            | $\hat{V}^s = \dot{g} g^{-1}$                                                                        |
| Adjoint                  | $V^s = \text{Ad}_g V^b, \text{Ad}_g \in \mathbb{R}^{6 \times 6}$                                    |
| Wrench pitch             | $h = f^T \tau / \ f\ ^2$                                                                            |
| POE forward kinematics   | $g(\theta) = e^{\hat{\xi}_1 \theta_1} \dots e^{\hat{\xi}_n \theta_n} g(0)$                          |
| Spatial Jacobian         | $V^s = J_s(\theta) \dot{\theta}$                                                                    |
| Joint torque from wrench | $\tau = J_s^T F_s = J_b^T F_b$ , frame 必须匹配                                                         |
| Kinetic energy           | $T = \frac{1}{2} \dot{q}^T M(q) \dot{q}$ , 一般依赖 $q$                                                 |
| Rigid body torque        | $\tau^b = I \dot{\omega}^b + \omega^b \times (I \omega^b)$                                          |
| Manipulator inertia      | $M(\theta)$ symmetric positive definite, 一般依赖 $\theta$                                              |
| Computed torque          | feedback linearization, 不是 workspace control 的必要核心                                                  |